

9.7 Polynomial Matrices and the Smith Normal Form

The rational canonical form of the previous section is a complete answer to the similarity problem, but the proof did not tell us how to *find* the invariant factors of a given matrix. This section supplies the missing algorithm, and it comes from an unexpected place: instead of working with A , we work with the single polynomial matrix $xI - A$, and we reduce it by row and column operations whose scalars are allowed to be polynomials. The reduced form is diagonal, its non-constant diagonal entries are the invariant factors, and two matrices are similar exactly when their reduced forms agree. On the way we finally pay off a promise from Chapter 3: every square matrix is similar to its transpose, over every field.

9.7.1 The space as a module over the polynomials

Everything in this chapter is built by applying **polynomials** in T to vectors: the T -annihilator of a vector, the cyclic subspace $Z(\mathbf{v}; T)$, the kernels $\ker p(T)$. Name that structure once. For V a vector space over F and $T \in \mathcal{L}(V)$, the product

$$F[x] \times V \rightarrow V, \quad p \cdot \mathbf{v} := p(T) \mathbf{v}$$

satisfies, by Theorem 5.7.1, exactly the vector space axioms with F replaced by $F[x]$: it is additive in each factor, $(pq) \cdot \mathbf{v} = p \cdot (q \cdot \mathbf{v})$, and $1 \cdot \mathbf{v} = \mathbf{v}$. A set with an addition and a product by the elements of a commutative ring R obeying these rules is an **R -module**, so the pair (V, T) is the same thing as an $F[x]$ -module that is finite-dimensional over the constants F . We develop no general theory of modules; the point is the dictionary:

$F[x]$ -module language	operator language
submodule of V	T -invariant subspace (Definition 8.1.1)
submodule generated by $\mathbf{v}$, that is $F[x] \cdot \mathbf{v}$	cyclic subspace $Z(\mathbf{v}; T)$ (Definition 9.5.1)
the annihilator of $\mathbf{v}$ in $F[x]$	the T -annihilator of $\mathbf{v}$
V is generated by one element	T has a cyclic vector

A subspace closed under multiplication by x is closed under multiplication by every polynomial, which is why the first two lines match.

Where does $xI - A$ come from here? Fix $\mathcal{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$ with $A = [T]_{\mathcal{B}}$. As an $F[x]$ -module, V is generated by $\mathbf{b}_1, \dots, \mathbf{b}_n$, subject to the n relations $x \cdot \mathbf{b}_j = \sum_i a_{ij} \mathbf{b}_i$, whose coefficient vectors are the columns of $xI - A$. So $xI - A$ is the **matrix of relations**, and changing generators or replacing one relation by a combination of relations should not change V . That is the intuition behind the operations we now allow; the proofs below are matrix proofs and do not rely on it.

9.7.2 Row and column operations with polynomial multipliers

A matrix in $M_n(F[x])$ will be called a **polynomial matrix**. Over a field we reduced a matrix by three kinds of row operation (Definition 2.2.1). Over $F[x]$ we must be careful with one of

them: multiplying a row by x cannot be undone, because x has no inverse in $F[x]$. So the scaling operation keeps its scalar in F , while the operation that adds a multiple of one row to another may use any polynomial multiplier.

Definition 9.7.1 — Elementary Operations over $F[x]$

Let $M \in M_n(F[x])$. An **elementary row operation** over $F[x]$ on M is one of:

- (E1) interchanging two rows;
- (E2) multiplying one row by a **non-zero scalar** $c \in F$;
- (E3) adding h times one row to a **different** row, where $h \in F[x]$ is **any** polynomial.

Elementary column operations over $F[x]$ are defined the same way with “column” in place of “row”.

In words: (E1) and (E3) are as over a field, except that (E3) now permits a polynomial multiplier; (E2) is restricted, because only the non-zero constants have inverses in $F[x]$. Each of the three is reversible by an operation of the same kind: swap back, scale by c^{-1} , add $-h$ times the row back.

Two polynomial matrices reachable from one another by such operations should count as the same. As over a field (Definition 3.8.1), the clean way to say this is with invertible matrices on both sides — but “invertible” must now mean invertible *inside* $M_n(F[x])$.

Proposition 9.7.1 — Invertible Polynomial Matrices

Let $U \in M_n(F[x])$. Then U has an inverse in $M_n(F[x])$ if and only if $\det U$ is a **non-zero constant**.

Proof. ($\Rightarrow$) Suppose $UW = I$ with $W \in M_n(F[x])$. Taking determinants and using multiplicativity over the commutative ring $F[x]$ (Theorem 6.5.4 and the remark on commutative rings after it), $\det U \cdot \det W = 1$. By Theorem 0.8.3, $\deg(\det U) + \deg(\det W) = \deg 1 = 0$, and degrees of non-zero polynomials are ≥ 0 , so $\deg(\det U) = 0$; that is, $\det U \in F \setminus \{0\}$.

($\Leftarrow$) Suppose $\det U = c \in F \setminus \{0\}$. The adjugate identity

$$U \operatorname{adj} U = \operatorname{adj} U U = (\det U) I$$

holds over any commutative ring (Theorem 6.6.2 and the remark on commutative rings following it), and $\operatorname{adj} U \in M_n(F[x])$, since its entries are minors of U . So $W = c^{-1} \operatorname{adj} U \in M_n(F[x])$ satisfies $UW = WU = I$.

■

Definition 9.7.2 — Equivalence over $F[x]$

Let $M, M' \in M_n(F[x])$. We say M and M' are **equivalent over** $F[x]$, written $M \approx M'$,

if there are $U, V \in M_n(F[x])$, **each invertible in** $M_n(F[x])$, with

$$M' = UMV.$$

Proposition 9.7.2 — Operations and Equivalence

Let $M, M' \in M_n(F[x])$.

- (a) $\approx$ is an equivalence relation on $M_n(F[x])$.
- (b) If M' is obtained from M by a finite sequence of elementary row and column operations over $F[x]$, then $M \approx M'$.

Proof.

- (a) Take $U = V = I$ for reflexivity. If $M' = UMV$, then $M = U^{-1}M'V^{-1}$, and U^{-1}, V^{-1} are invertible in $M_n(F[x])$; this gives symmetry. If $M' = UMV$ and $M'' = U'M'V'$, then $M'' = (U'U)M(VV')$, and a product of invertible matrices is invertible, which gives transitivity.
- (b) Each elementary row operation over $F[x]$ is left multiplication by the matrix obtained by performing that operation on I_n , exactly as over a field (Theorem 2.4.1, whose proof uses only the ring operations). Those three matrices are P_{ij} , $D_i(c)$ and $I + hE_{ij}$ with $i \neq j$, of determinants -1 , $c \neq 0$ and 1 respectively (Theorem 6.4.6 and Theorem 6.5.1, valid over a commutative ring), hence invertible in $M_n(F[x])$ by Proposition 9.7.1. Column operations are right multiplication by the same matrices. Now use (a).

■

Warning. Multiplying a row by a non-constant polynomial is not an allowed operation. It changes the answer. The matrix $M = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ becomes $\begin{pmatrix} x & 0 \\ 0 & 1 \end{pmatrix}$ after “multiplying row 1 by x ”, and the two are not equivalent: their determinants are 1 and x , and by Proposition 9.7.1 the first is invertible over $F[x]$ while the second is not. The rule to remember is that only the **units** of $F[x]$ — the non-zero constants — may be used as scaling factors.

9.7.3 The Smith normal form

Every polynomial matrix can be brought to a diagonal form whose diagonal entries divide one another in turn, and that form is unique.

The existence proof is Euclidean division, run on whichever non-zero entry has the smallest degree: if that entry fails to divide one of its neighbors in the same row or column, the remainder gives a non-zero entry of strictly smaller degree, and degrees cannot decrease forever.

Theorem 9.7.3 — Smith Normal Form

Let F be a field, $n \geq 1$ and $M \in M_n(F[x])$. Then:

- (a) **(Existence)** M is equivalent over $F[x]$, by a finite sequence of elementary row and column operations, to a diagonal matrix

$$D = \text{diag}(d_1, d_2, \dots, d_n)$$

in which every d_i is **monic or zero** and $d_i \mid d_{i+1}$ for $i = 1, \dots, n-1$.

- (b) **(Uniqueness)** The list $(d_1, \dots, d_n)$ is determined by M : if $M \approx \text{diag}(d_1, \dots, d_n) \approx \text{diag}(e_1, \dots, e_n)$ with both lists as in (a), then $d_i = e_i$ for every i .

The matrix D is called the **Smith normal form** of M , and $d_1, \dots, d_n$ its **invariant divisors**.

Since $0 \mid f$ holds only for $f = 0$, the divisibility chain forces the zero entries, if any, to be the last ones.

Idea. Step roadmap for existence. ① Among all matrices reachable from M by operations, pick one whose smallest non-zero entry degree is least, and move that entry to the corner. ② That corner entry f must divide every other entry of its row and column: otherwise divide with remainder and produce an entry of smaller degree, contradicting the choice in ①. ③ Clear the first row and column with f ; the same remainder trick, after adding a row to the first one, shows f divides every entry of what is left. ④ Induct on the size. The divisibility $d_1 \mid d_2$ survives the induction because operations replace entries by $F[x]$ -combinations of entries.

For uniqueness, find a quantity that can be read off both M and D . The greatest common divisor of all $k \times k$ minors is such a quantity: minors of a product are $F[x]$ -combinations of minors of the factors (Cauchy–Binet), so the gcd cannot change under equivalence, and for a diagonal matrix with a divisibility chain it is simply $d_1 d_2 \cdots d_k$.

Proof.

- (a) If $M = 0$, it is already of the stated form. So assume $M \neq 0$.
For a non-zero $N \in M_n(F[x])$ put

$$\delta(N) := \min\{\deg n_{ij} : n_{ij} \neq 0\} \in \mathbb{N},$$

the least degree of a non-zero entry. Let $\mathcal{S}$ be the set of all non-zero matrices obtainable from M by finite sequences of elementary row and column operations over $F[x]$; it contains M . The set $\{\delta(N) : N \in \mathcal{S}\}$ is a non-empty subset of $\mathbb{N}$, so it has a least element; **fix** $N \in \mathcal{S}$ **with** $\delta(N)$ **least**.

Step 1. Put a least-degree entry in the corner, monic. Choose (i, j) with $n_{ij} \neq 0$ and $\deg n_{ij} = \delta(N)$. Swapping row i with row 1 and column j with column 1 (operations (E1)), and then scaling the new first row by the inverse of the leading coefficient of the corner entry (operation (E2)), we obtain a matrix in $\mathcal{S}$ whose $(1, 1)$ -entry f is **monic** of degree $\delta(N)$. These operations permute the entries and rescale some of them, so the new matrix still has least non-zero entry degree $\delta(N)$. Rename it N .

Step 2. f divides every entry of the first row and the first column. Suppose $f \nmid n_{1j}$ for some $j \geq 2$. By Theorem 5.1.1, $n_{1j} = qf + r$ with $r \neq 0$ and $\deg r < \deg f$. Adding $-q$ times column 1 to column j (operation (E3)) produces a matrix $N' \in \mathcal{S}$ whose $(1, j)$ -entry is $r \neq 0$ with $\deg r < \deg f = \delta(N)$. Then

$\delta(N') < \delta(N)$, contradicting the minimality of $\delta(N)$. Hence $f \mid n_{1j}$ for every j , and the same argument with rows in place of columns gives $f \mid n_{i1}$ for every i .

Step 3. Clear the first row and column. For $j \geq 2$ add $-(n_{1j}/f)$ times column 1 to column j ; the multipliers are polynomials by Step 2, and after these operations the first row is $(f, 0, \dots, 0)$ while the first column is unchanged. Then for $i \geq 2$ add $-(n_{i1}/f)$ times row 1 to row i ; since the first row is now $(f, 0, \dots, 0)$, this clears the first column and changes nothing else. The result is

$$\begin{pmatrix} f & 0 \\ 0 & M' \end{pmatrix} \in \mathcal{S}, \quad M' \in M_{n-1}(F[x]),$$

and its least non-zero entry degree is still $\delta(N)$: it is $\geq \delta(N)$ by minimality, and f is an entry of degree $\delta(N)$.

Step 4. f divides every entry of M' . Suppose $f \nmid (M')_{kl}$ for some k, l . Add row $k+1$ to row 1. The new first row is $(f, (M')_{k1}, \dots, (M')_{k,n-1})$, so the $(1, l+1)$ -entry of the resulting matrix $N'' \in \mathcal{S}$ is $(M')_{kl}$, which is not divisible by f , while the $(1, 1)$ -entry is still f , monic of degree $\delta(N)$. By minimality $\delta(N'') = \delta(N)$, so Step 2 applies to N'' and says that f divides every entry of the first row of N'' — a contradiction.

Step 5. Induct on n . We prove (a) by induction on n . For $n = 1$, scale the single entry to be monic (or leave it 0). For $n \geq 2$, Steps 1–4 give a sequence of operations taking M to $f \oplus M'$ with f monic and f dividing every entry of M' . By the induction hypothesis there are operations taking M' to $D' = \text{diag}(d_2, \dots, d_n)$ with every d_i monic or zero and $d_2 \mid \dots \mid d_n$; performing the corresponding operations on rows and columns 2, $\dots$, n of $f \oplus M'$ takes it to $f \oplus D' = \text{diag}(f, d_2, \dots, d_n)$. It remains to see that $f \mid d_2$. By Proposition 9.7.2, $D' = U'M'V'$ for some $U', V' \in M_{n-1}(F[x])$, so every entry of D' is a sum of products $u'_{ik}(M')_{kl}v'_{lj}$ (Definition 0.9.9) and is therefore divisible by f , since every $(M')_{kl}$ is. In particular $f \mid d_2$. Setting $d_1 = f$ finishes the induction.

- (b) Call a set $I \subseteq [n]$ with $|I| = k$ a **k -set**. For $M \in M_n(F[x])$ and $1 \leq k \leq n$ define the k -th **determinantal divisor**

$$\Delta_k(M) := \text{the monic generator of the ideal } \langle \det M_{I,J} : I, J \text{ } k\text{-sets} \rangle \subseteq F[x],$$

and $\Delta_k(M) := 0$ if all the $k \times k$ minors of M vanish. This is well defined by Proposition 5.2.1 and Theorem 5.2.2: the ideal generated by the finitely many $k \times k$ minors is $\langle \Delta_k(M) \rangle$ for exactly one monic $\Delta_k(M)$ unless the ideal is $\{0\}$. Equivalently, $\Delta_k(M)$ is the monic greatest common divisor of the $k \times k$ minors: it divides each of them, and every common divisor of them divides it (Theorem 5.2.3, extended from two polynomials to finitely many by the same ideal argument).

Claim 1. $M \approx M'$ implies $\Delta_k(M) = \Delta_k(M')$ for every k .

Proof. Let $M' = UMV$ with U, V invertible over $F[x]$, and let I, J be k -sets. The submatrix of a product on rows I and columns J is the product of the submatrix of the left factor on rows I and all columns with the submatrix of the right factor on all rows and columns J (Definition 0.9.9, read off entry by entry). Applying this twice and then Theorem 6.7.2 over the commutative ring $F[x]$ — legitimate by the remark after that theorem — to the $k \times n$ times $n \times k$ products involved,

$$\det M'_{I,J} = \sum_S \det U_{I,S} \det (MV)_{S,J} = \sum_S \sum_R \det U_{I,S} \det M_{S,R} \det V_{R,J},$$

with S and R running over the k -sets. So every $k \times k$ minor of M' is an $F[x]$ -combination of $k \times k$ minors of M , hence lies in $\langle \det M_{S,R} \rangle$. Therefore $\Delta_k(M)$ divides every $k \times k$ minor of M' , and so $\Delta_k(M) \mid \Delta_k(M')$. Exchanging the roles of M and M' using $M = U^{-1}M'V^{-1}$ gives $\Delta_k(M') \mid \Delta_k(M)$. Two monic polynomials dividing each other are equal (Proposition 5.1.2), and if one is 0 so is the other. ■

Claim 2. If $D = \text{diag}(d_1, \dots, d_n)$ with $d_1 \mid d_2 \mid \dots \mid d_n$, then $\Delta_k(D) = d_1 d_2 \cdots d_k$.

Proof. Let I, J be k -sets. If $I \neq J$, pick $i \in I \setminus J$; row i of D has its only possibly non-zero entry in column $i \notin J$, so the submatrix $D_{I,J}$ has a zero row and $\det D_{I,J} = 0$. If $I = J = \{i_1 < \dots < i_k\}$, then $D_{I,I} = \text{diag}(d_{i_1}, \dots, d_{i_k})$ and $\det D_{I,I} = d_{i_1} \cdots d_{i_k}$ (Theorem 6.4.6, valid over the commutative ring $F[x]$ by the remark after it). Since $i_j \geq j$, the chain gives $d_j \mid d_{i_j}$ for each j , hence

$$d_1 d_2 \cdots d_k \mid d_{i_1} d_{i_2} \cdots d_{i_k}.$$

So $d_1 \cdots d_k$ divides every $k \times k$ minor of D , and it is itself the minor for $I = J = \{1, \dots, k\}$. Hence the ideal generated by the $k \times k$ minors is $\langle d_1 \cdots d_k \rangle$. If $d_1 \cdots d_k \neq 0$ it is monic, being a product of monic polynomials, and equals $\Delta_k(D)$. If $d_1 \cdots d_k = 0$, say $d_m = 0$ with $m \leq k$, then for every k -set I we have $i_k \geq k \geq m$, so $d_m \mid d_{i_k}$ forces $d_{i_k} = 0$ and the minor vanishes; thus $\Delta_k(D) = 0 = d_1 \cdots d_k$. ■

Now let $M \approx \text{diag}(d_1, \dots, d_n)$ with the d_i as in (a). By Claims 1 and 2,

$$\Delta_k(M) = d_1 d_2 \cdots d_k \quad (k = 1, \dots, n),$$

a quantity depending only on M . Reading this backwards recovers the d_i : first $d_1 = \Delta_1(M)$; and for $k \geq 2$, if $\Delta_{k-1}(M) \neq 0$ then $d_k = \Delta_k(M)/\Delta_{k-1}(M)$, while if $\Delta_{k-1}(M) = 0$ then $d_1 \cdots d_{k-1} = 0$, so some $d_m = 0$ with $m \leq k-1$, and the chain forces $d_k = 0$. Hence any second list $(e_1, \dots, e_n)$ as in (a) satisfies the same recipe and equals $(d_1, \dots, d_n)$. This proves the theorem. ■

The proof is an algorithm, and it is the one to use by hand: hunt for the entry of least degree,

divide, and repeat.

Smith normal form of $M \in M_n(F[x])$.

1. If $M = 0$, stop. Otherwise find a non-zero entry of least degree and move it to position $(1, 1)$ by a row and a column swap; scale row 1 so that the entry is monic. Call it f .
2. For each other entry g in row 1 or column 1 with $f \nmid g$, replace g by its remainder on division by f (subtract a multiple of column 1, resp. row 1), move the remainder to the corner, and go back to step 1.
3. When f divides all of row 1 and column 1, clear them. If some entry g of the remaining block is not divisible by f , add its row to row 1 and go back to step 2.
4. Repeat on the remaining $(n - 1) \times (n - 1)$ block.
5. Check the answer: Δ_k must equal $d_1 \cdots d_k$. In particular $d_1 \cdots d_n$ is $\det M$ rescaled by a non-zero constant to be monic, and d_1 is the monic gcd of the entries.

Quick check. Two polynomial matrices in $M_2(F[x])$ have determinants x^2 and x^2 , and in both the greatest common divisor of the four entries is 1. Are they equivalent over $F[x]$?

Solution. Yes. Here $\Delta_1 = 1$ (the gcd of the 1×1 minors, that is, of the entries) and Δ_2 is the determinant rescaled to be monic, so $\Delta_2 = x^2$ for both. By [Theorem 9.7.3](#) both have Smith normal form $\text{diag}(1, x^2)$, and since $\approx$ is an equivalence relation ([Proposition 9.7.2 \(a\)](#)) they are equivalent to each other. For 2×2 matrices the pair (Δ_1, Δ_2) is therefore a complete invariant.

Example. (A Smith Normal Form Computed by Hand)

Let

$$A = \begin{pmatrix} 2 & 1 & -2 \\ 1 & 2 & -2 \\ 1 & 1 & -1 \end{pmatrix} \in M_3(\mathbb{Q}).$$

Find the Smith normal form of $xI - A$ over $\mathbb{Q}[x]$, naming every operation, and check the answer with determinantal divisors.

Solution. We have

$$xI - A = \begin{pmatrix} x-2 & -1 & 2 \\ -1 & x-2 & 2 \\ -1 & -1 & x+1 \end{pmatrix}.$$

Find the corner. The entries of least degree are the three entries equal to -1 . Swap columns 1 and 2, then multiply the first row by -1 :

$$\begin{pmatrix} -1 & x-2 & 2 \\ x-2 & -1 & 2 \\ -1 & -1 & x+1 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & -(x-2) & -2 \\ x-2 & -1 & 2 \\ -1 & -1 & x+1 \end{pmatrix}.$$

The corner is now 1, which divides everything, so no division step is needed.

Clear the first column. Add $-(x-2)$ times row 1 to row 2, and add row 1 to row 3. In row 2 this gives entries $-1 + (x-2)^2 = (x-1)(x-3)$ and $2 + 2(x-2) = 2(x-1)$; in

row 3 it gives $-1 - (x - 2) = -(x - 1)$ and $x + 1 - 2 = x - 1$:

$$\begin{pmatrix} 1 & -(x-2) & -2 \\ 0 & (x-1)(x-3) & 2(x-1) \\ 0 & -(x-1) & x-1 \end{pmatrix}.$$

Clear the first row. Add $(x - 2)$ times column 1 to column 2, and 2 times column 1 to column 3. Since the first column below the corner is zero, nothing else changes:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & (x-1)(x-3) & 2(x-1) \\ 0 & -(x-1) & x-1 \end{pmatrix} = \text{diag}(1) \oplus \begin{pmatrix} (x-1)(x-3) & 2(x-1) \\ -(x-1) & x-1 \end{pmatrix}.$$

Every entry of the 2×2 block is divisible by 1, so we move on to the block.

The block. Its entries of least degree are the three of degree 1. Swap rows 2 and 3 and scale the new row 2 by -1 , putting the monic $x - 1$ in the corner of the block:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & x-1 & -(x-1) \\ 0 & (x-1)(x-3) & 2(x-1) \end{pmatrix}.$$

Now $x - 1$ divides every entry of its row and column. Add column 2 to column 3, which clears the block's first row and turns the $(3, 3)$ -entry into $2(x - 1) + (x - 1)(x - 3) = (x - 1)(x - 1)$:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & x-1 & 0 \\ 0 & (x-1)(x-3) & (x-1)^2 \end{pmatrix}.$$

Add $-(x - 3)$ times row 2 to row 3 to clear the block's first column:

$$D = \text{diag}(1, x - 1, (x - 1)^2),$$

and $1 \mid x - 1 \mid (x - 1)^2$, so this is the Smith normal form.

The check. The matrix $xI - A$ has an entry -1 , so $\Delta_1 = 1$. Its determinant is $p_A = (x - 1)^3$ — expanding along the first row,

$$(x-2)[(x-2)(x+1)+2] + [-(x+1)+2] + 2[1+(x-2)] = (x-2)(x^2-x) + (1-x) + 2(x-1),$$

which is $(x-2)x(x-1) + (x-1) = (x-1)(x^2-2x+1) = (x-1)^3$ — so $\Delta_3 = (x-1)^3$. For Δ_2 , the minor on rows $\{2, 3\}$ and columns $\{1, 2\}$ is $\det \begin{pmatrix} -1 & x-2 \\ -1 & -1 \end{pmatrix} = 1 + (x-2) = x-1$, so $\Delta_2 \mid x-1$; and every 2×2 minor is divisible by $x-1$, as Claim 2 predicts from D . Hence $\Delta_2 = x-1$, and

$$(d_1, d_2, d_3) = \left(\Delta_1, \frac{\Delta_2}{\Delta_1}, \frac{\Delta_3}{\Delta_2} \right) = (1, x - 1, (x - 1)^2),$$

in agreement. As a sanity check on A itself: $A - I$ has rank 1, so $g_A(1) = 2$ and the Jordan form of A is $J_2(1) \oplus J_1(1)$ (Corollary 9.3.2).

9.7.4 Similarity read off from a polynomial matrix

We can now state the theorem that makes all of this useful. It converts a question about A and B — is there an invertible $P \in M_n(F)$ with $B = P^{-1}AP$? — into a question about the two polynomial matrices $xI - A$ and $xI - B$, which the Smith algorithm answers by computation.

One direction is immediate. The other needs a tool: polynomial matrices can be divided by $xI - A$, with a **constant** remainder.

Lemma 9.7.4 — Division of a Polynomial Matrix by $xI - A$

Let $A \in M_n(F)$ and $M \in M_n(F[x])$. Then there are $Q \in M_n(F[x])$ and $R \in M_n(F)$ with

$$M = (xI - A)Q + R.$$

Idea. Write M as a polynomial $\sum_k M_k x^k$ with **constant matrix** coefficients. The scalar identity $x^k - a^k = (x - a)(x^{k-1} + \dots + a^{k-1})$ has a matrix version $x^k I - A^k = (xI - A)S_k$, in which the second factor is a polynomial in x and A and so commutes with $xI - A$. Replacing each $x^k I$ by $(x^k I - A^k) + A^k$ puts a factor $xI - A$ in front of everything except a constant matrix.

Proof. Collecting the coefficients of each power of x in the entries of M , write $M = \sum_{k=0}^m x^k M_k$ with $M_0, \dots, M_m \in M_n(F)$; note that x^k is a scalar polynomial, so $x^k M_k = M_k x^k$. For $k \geq 1$ put

$$S_k := \sum_{j=0}^{k-1} x^{k-1-j} A^j \in M_n(F[x]), \quad S_0 := 0.$$

Expanding the product and telescoping,

$$(xI - A)S_k = \sum_{j=0}^{k-1} x^{k-j} A^j - \sum_{j=0}^{k-1} x^{k-1-j} A^{j+1} = x^k I - A^k$$

for every $k \geq 0$. Therefore

$$M = \sum_{k=0}^m (x^k I) M_k = \sum_{k=0}^m (x^k I - A^k) M_k + \sum_{k=0}^m A^k M_k = (xI - A) \underbrace{\sum_{k=0}^m S_k M_k}_{=Q} + \underbrace{\sum_{k=0}^m A^k M_k}_{=R},$$

with $Q \in M_n(F[x])$ and $R \in M_n(F)$, as claimed. ■

Theorem 9.7.5 — Similarity as Equivalence of $xI - A$ and $xI - B$

Let F be a field, $n \geq 1$ and $A, B \in M_n(F)$. Then

$$A \sim B \text{ over } F \quad \Longleftrightarrow \quad xI - A \approx xI - B \text{ over } F[x].$$

Idea. $(\Rightarrow)$ is a one-line computation: a constant invertible P is also an invertible polynomial matrix.

$(\Leftarrow)$ is the substance, and the whole difficulty is that the matrices U, V witnessing the equivalence have polynomial entries, while we must produce a **constant** P . The trick is to divide U by $xI - B$ and keep only the constant remainder P . Comparing degrees in the resulting identity shows that P alone already conjugates: $P(xI - A) = (xI - B)P$, that is, $PA = BP$. Running the same argument on the inverted relation produces a second constant P' with $P'B = AP'$, and multiplying the two expressions for V^{-1} and V shows $PP' = I$, so P is invertible.

Proof. $(\Rightarrow)$ Suppose $B = P^{-1}AP$ with $P \in M_n(F)$ invertible. Then, inside $M_n(F[x])$,

$$xI - B = P^{-1}(xI)P - P^{-1}AP = P^{-1}(xI - A)P,$$

and $P^{-1}, P \in M_n(F) \subseteq M_n(F[x])$ have non-zero constant determinants, hence are invertible over $F[x]$ (Proposition 9.7.1). So $xI - A \approx xI - B$.

$(\Leftarrow)$ Suppose

$$xI - B = U(xI - A)V \quad (*)$$

with $U, V \in M_n(F[x])$ invertible over $F[x]$.

Step 1. Divide U by $xI - B$. By Lemma 9.7.4 there are $Q \in M_n(F[x])$ and $P \in M_n(F)$ with $U = (xI - B)Q + P$. Substituting into $(*)$,

$$xI - B = (xI - B)Q(xI - A)V + P(xI - A)V,$$

so $P(xI - A)V = (xI - B)[I - Q(xI - A)V]$. Multiplying on the right by V^{-1} and putting

$$S := V^{-1} - Q(xI - A) \in M_n(F[x]),$$

we get

$$P(xI - A) = (xI - B)S. \quad (1)$$

Step 2. S is a constant matrix, and $PA = BP$. Suppose $S \neq 0$ and write $S = \sum_{k=0}^d x^k S_k$ with $S_k \in M_n(F)$ and $S_d \neq 0$. The coefficient of x^{d+1} in $(xI - B)S$ is $S_d \neq 0$, while the left-hand side of (1) is $xP - PA$, whose coefficients vanish beyond x^1 . Hence $d+1 \leq 1$, that is $d = 0$. So in every case $S \in M_n(F)$. Comparing the coefficients of x^1 in (1) gives $P = S$, and comparing the constant terms gives $-PA = -BP$. Therefore

$$PA = BP, \quad \text{equivalently} \quad P(xI - A) = (xI - B)P, \quad (2)$$

the second form because $P(xI - A) = xP - PA = xP - BP = (xI - B)P$. Moreover $S = P$ turns the definition of S into

$$V^{-1} = P + Q(xI - A). \quad (3)$$

Step 3. The same argument, inverted. From $(*)$, $xI - A = U^{-1}(xI - B)V^{-1}$, which is $(*)$ with A, B interchanged and U, V replaced by the invertible polynomial matrices U^{-1}, V^{-1} . So Steps 1 and 2 apply and give $Q' \in M_n(F[x])$ and $P' \in M_n(F)$ with

$$P'(xI - B) = (xI - A)P' \quad \text{and} \quad V = P' + Q'(xI - B). \quad (4)$$

Step 4. P is invertible. Multiplying (3) by (4),

$$I = V^{-1}V = [P+Q(xI-A)][P'+Q'(xI-B)] = PP' + PQ'(xI-B) + Q(xI-A)P' + Q(xI-A)Q'(xI-B).$$

By the first identity in (4), $(xI-A)P' = P'(xI-B)$, so the third summand is $QP'(xI-B)$ and

$$I - PP' = Y(xI - B), \quad Y := PQ' + QP' + Q(xI - A)Q' \in M_n(F[x]).$$

If $Y \neq 0$, write $Y = \sum_{k=0}^e x^k Y_k$ with $Y_e \neq 0$; the coefficient of x^{e+1} on the right is $Y_e \neq 0$, while the left-hand side is constant. Hence $Y = 0$ and $PP' = I$. By Theorem 2.4.5, P is invertible with $P^{-1} = P'$.

Step 5. Conclude. By (2), $PA = BP$, so $B = PAP^{-1}$ and $A \sim B$. This proves the theorem. ■

Two by-products are worth recording. Whenever $xI - B = U(xI - A)V$ with U, V invertible over $F[x]$, taking determinants gives $p_B = (\det U)(\det V) p_A$ with $\det U, \det V$ non-zero constants (Proposition 9.7.1); both p_A and p_B are monic of degree n , so

$$p_A = p_B \quad \text{and} \quad (\det U)(\det V) = 1,$$

which recovers the invariance of the characteristic polynomial. And, by Theorem 9.7.3, the test is now decidable by a finite computation in $F[x]$: reduce $xI - A$ and $xI - B$ to Smith normal form and compare.

Corollary 9.7.6 — The Smith Form of $xI - A$ Carries the Invariant Factors

Let $A \in M_n(F)$ with invariant factors $d_1 \mid d_2 \mid \cdots \mid d_r$ as in Theorem 9.6.5. Then the Smith normal form of $xI - A$ is

$$\text{diag} \left(\underbrace{1, \dots, 1}_{n-r}, d_1, d_2, \dots, d_r \right).$$

In particular the invariant factors of A are exactly the non-constant diagonal entries of the Smith normal form of $xI - A$, read from the top down, and two matrices in $M_n(F)$ are similar if and only if $xI - A$ and $xI - B$ have the same Smith normal form.

Proof. By Theorem 9.6.5, $A \sim C(d_1) \oplus \cdots \oplus C(d_r)$ with d_i monic, $\deg d_i = k_i \geq 1$ and $\sum k_i = n$. By the ($\Rightarrow$) direction of Theorem 9.7.5 it is enough to find the Smith normal form of

$$xI_n - (C(d_1) \oplus \cdots \oplus C(d_r)) = (xI_{k_1} - C(d_1)) \oplus \cdots \oplus (xI_{k_r} - C(d_r)),$$

the equality by Theorem 7.1.4 (a).

One block. Let d be monic of degree $k \geq 1$. By Definition 9.5.3, the only non-zero entries of $xI_k - C(d)$ are x on the diagonal (with $x + a_{k-1}$ in the corner), -1 in each position $(i+1, i)$, and the entries of the last column. Delete row 1 and column k . In the remaining $(k-1) \times (k-1)$ matrix, the entry in position (p, q) is the $(p+1, q)$ -entry of $xI_k - C(d)$: it is -1 when $q = p$, it is x when $q = p+1$, and it is 0 when $q < p$.

So that matrix is upper triangular with every diagonal entry -1 , and its determinant is $(-1)^{k-1} \neq 0$ by Theorem 6.4.6, again over $F[x]$. Hence $\Delta_{k-1}(xI_k - C(d)) = 1$. By Claim 2 in the proof of Theorem 9.7.3, the determinantal divisors of any matrix are the partial products $d_1 \cdots d_j$ of its Smith diagonal, so $\Delta_j \mid \Delta_{k-1} = 1$ and therefore $\Delta_j = 1$ for every $j \leq k-1$. Also $\Delta_k = \det(xI_k - C(d)) = p_{C(d)} = d$ by Theorem 9.5.3. By the uniqueness part of Theorem 9.7.3, the Smith normal form of $xI_k - C(d)$ is $\text{diag}(1, \dots, 1, d)$. (For $k = 1$ read this as $\text{diag}(d)$.)

Assembling. If $M_i \approx M'_i$ with witnesses U_i, V_i , then $\bigoplus_i M_i \approx \bigoplus_i M'_i$ with witnesses $\bigoplus_i U_i$ and $\bigoplus_i V_i$, which are invertible with inverses $\bigoplus_i U_i^{-1}$, $\bigoplus_i V_i^{-1}$ (Theorem 7.1.4 (a), (c)). So $xI_n - A$ is equivalent to the direct sum of the matrices $\text{diag}(1, \dots, 1, d_i)$, a diagonal matrix whose diagonal entries are $n - r$ ones and the polynomials $d_1, \dots, d_r$. Simultaneous row and column swaps — each a pair of elementary operations — reorder these entries at will, so $xI_n - A$ is equivalent to

$$\text{diag}(1, \dots, 1, d_1, d_2, \dots, d_r),$$

and this is already in the shape required by Theorem 9.7.3 (a), because $1 \mid d_1 \mid d_2 \mid \cdots \mid d_r$ and all the entries are monic. By uniqueness it *is* the Smith normal form of $xI - A$. The final claim combines this with Theorem 9.7.5 and the uniqueness in Theorem 9.7.3. ■

So the determinantal divisors of $xI - A$ compute the invariant factors of A without any factorization of polynomials: Δ_k is a gcd, obtained from the Euclidean algorithm alone. Moreover $\Delta_n(xI - A) = p_A$ and $\Delta_n/\Delta_{n-1} = d_r = m_A$.

9.7.5 Every matrix is similar to its transpose

Chapter 3 asked whether A and $A^\top$ are always similar and postponed the answer; Exercise (C1: A matrix is similar to its transpose) settled it whenever p_A splits. The Smith form removes that hypothesis, because transposing a matrix does not change its minors.

Corollary 9.7.7 — A Matrix Is Similar to Its Transpose

Let F be any field and $A \in M_n(F)$. Then $A \sim A^\top$.

Proof. Let $M = xI - A$, so that $M^\top = xI - A^\top$. For k -sets I, J we have $(M^\top)_{I,J} = (M_{J,I})^\top$ by Definition 0.9.10, so $\det(M^\top)_{I,J} = \det M_{J,I}$ by Theorem 6.4.5, valid over the commutative ring $F[x]$. Hence the $k \times k$ minors of $M^\top$ are exactly the $k \times k$ minors of M , listed in a different order, and

$$\Delta_k(xI - A^\top) = \Delta_k(xI - A) \quad (k = 1, \dots, n).$$

By Theorem 9.7.3, each of $xI - A$ and $xI - A^\top$ is equivalent over $F[x]$ to the diagonal matrix built from its own determinantal divisors, and these two diagonal matrices coincide. Since $\approx$ is an equivalence relation (Proposition 9.7.2 (a)), $xI - A \approx xI - A^\top$. By Theorem 9.7.5, $A \sim A^\top$. ■

Two sentences of consequence. An operator and its dual have the same invariants: by Theorem 4.4.2 the matrix of T' in the dual basis is $[T]_B^T$, so T and T' share their characteristic polynomial, their minimal polynomial and their invariant factors. And a matrix and its transpose always have the same Jordan form when one exists, which says that the *rows* of $A - \lambda I$ are exactly as informative as its columns.

Warning. **Similar is not the same as “equivalent over $F[x]$ ” for the matrices themselves.** The theorem compares $xI - A$ with $xI - B$, not A with B . As polynomial matrices, any two invertible $A, B \in M_n(F)$ are equivalent over $F[x]$ — take $U = A^{-1}$, $V = B$ — whether or not they are similar. The whole content of Theorem 9.7.5 is that inserting the variable x on the diagonal remembers the conjugation.

9.7.6 Exercises

9.7.6.1 A. Check your understanding

Exercise A1.

- List the three elementary operations over $F[x]$, and say which one differs from its counterpart over a field and why.
- State a criterion for a matrix $U \in M_n(F[x])$ to be invertible in $M_n(F[x])$.
- State the Smith Normal Form theorem, including the uniqueness clause.
- Define the k -th determinantal divisor of a polynomial matrix, and say why it is unchanged by equivalence.
- True or false: if $\det(xI - A) = \det(xI - B)$, then $xI - A \approx xI - B$. Justify your answer.
- How do you read m_A off the Smith normal form of $xI - A$?

Solution.

- Swap two rows; multiply a row by a non-zero **scalar** of F ; add a **polynomial** multiple of one row to a different row (Definition 9.7.1). The scaling operation is restricted, because only the non-zero constants are invertible in $F[x]$; multiplying a row by x is not reversible.
- U is invertible over $F[x]$ if and only if $\det U$ is a non-zero constant (Proposition 9.7.1).
- Every $M \in M_n(F[x])$ is equivalent over $F[x]$, by elementary operations, to a diagonal $\text{diag}(d_1, \dots, d_n)$ with each d_i monic or zero and $d_i \mid d_{i+1}$; this diagonal is unique (Theorem 9.7.3).
- $\Delta_k(M)$ is the monic generator of the ideal generated by all $k \times k$ minors of M , equivalently their monic gcd. It is unchanged by equivalence because every $k \times k$ minor of UMV is an $F[x]$ -combination of $k \times k$ minors of M , by the Cauchy–Binet formula (Claim 1 in the proof of Theorem 9.7.3).
- False. Equality of determinants says only $\Delta_n(xI - A) = \Delta_n(xI - B)$; the lower determinantal divisors can differ. For $A = J_2(0) \oplus J_2(0)$ and $B = J_3(0) \oplus J_1(0)$ in $M_4(F)$, both determinants are x^4 , but the Smith normal forms are $\text{diag}(1, 1, x^2, x^2)$ and $\text{diag}(1, 1, x, x^3)$. (Equivalently, the matrices are not similar, by Corollary 9.3.3.)

- (f) m_A is the **last** diagonal entry of the Smith normal form of $xI - A$, that is $\Delta_n/\Delta_{n-1} = p_A/\Delta_{n-1}$ (Corollary 9.7.6, since the last diagonal entry is $d_r = m_A$ in the notation there).

9.7.6.2 B. Practice

Exercise B1: Smith normal forms.

Find the Smith normal form over $\mathbb{Q}[x]$ of each of the following, naming the operations you use.

- (a) $M_1 = \begin{pmatrix} x & x+1 \\ x-1 & x \end{pmatrix}.$
- (b) $M_2 = \begin{pmatrix} x^2-1 & 0 \\ 0 & x^2-x \end{pmatrix}.$
- (c) $M_3 = \begin{pmatrix} x & 0 & 1 \\ -1 & x & 0 \\ 0 & -1 & x \end{pmatrix}.$

Solution.

- (a) $\det M_1 = x^2 - (x+1)(x-1) = 1$, so M_1 is invertible over $\mathbb{Q}[x]$ (Proposition 9.7.1) and $\Delta_1 = \Delta_2 = 1$. The Smith normal form is $\text{diag}(1, 1) = I_2$. (Concretely: subtract column 1 from column 2 to get $\begin{pmatrix} x & 1 \\ x-1 & 1 \end{pmatrix}$, then subtract row 1 from row 2, swap columns, and clear.)
- (b) Here $\Delta_1 = \gcd(x^2-1, x^2-x) = \gcd((x-1)(x+1), x(x-1)) = x-1$ and $\Delta_2 = \det M_2 = x(x-1)^2(x+1)$. So the Smith normal form is

$$\text{diag}\left(x-1, \frac{x(x-1)^2(x+1)}{x-1}\right) = \text{diag}(x-1, x(x-1)(x+1)),$$

and indeed $x-1$ divides $x(x-1)(x+1)$. By operations: add column 2 to column 1, giving first column (x^2-1, x^2-x) ; subtract row 1 from row 2 to get $(x^2-1, 1-x)$ in column 1; the entry $1-x$ has least degree, so swap it into the corner, scale by -1 , and continue.

- (c) The 2×2 minor on rows $\{2, 3\}$ and columns $\{1, 2\}$ is $\det \begin{pmatrix} -1 & x \\ 0 & -1 \end{pmatrix} = 1$, so $\Delta_2 = 1$, and then $\Delta_1 = 1$ as well (it divides Δ_2). Expanding $\det M_3$ along the first row, $x(x^2-0) + 1 \cdot (1-0) = x^3+1$, so $\Delta_3 = x^3+1$ and the Smith normal form is $\text{diag}(1, 1, x^3+1)$. Note $M_3 = xI - C$ with $C = \begin{pmatrix} 0 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = C(x^3+1)$, and Corollary 9.7.6 predicts exactly this, since a companion matrix has the single invariant factor x^3+1 .

Exercise B2: Invariant factors from determinantal divisors.

For each matrix A , compute $\Delta_1, \dots, \Delta_n$ of $xI - A$, write down the Smith normal form, and read off the invariant factors, m_A and the Jordan form.

- (a) $A = J_2(0) \oplus J_2(0) \in M_4(\mathbb{Q})$.
 (b) $A = J_3(0) \oplus J_1(0) \in M_4(\mathbb{Q})$.
 (c) $A = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix} \in M_2(\mathbb{Q})$.

Solution. One remark makes the minors easy. If $M = M_1 \oplus M_2$ with square blocks, and I, J are k -sets, split $I = I_1 \cup I_2$ and $J = J_1 \cup J_2$ according to the blocks. If $|I_1| \neq |J_1|$, then $M_{I,J}$ has a zero row or a zero column and $\det M_{I,J} = 0$; if $|I_1| = |J_1|$, then $M_{I,J} = (M_1)_{I_1,J_1} \oplus (M_2)_{I_2,J_2}$ and the determinant is the product of the two, by Theorem 6.8.4, whose proof uses only sums and products of entries and so applies over $F[x]$ (the remark after that theorem). So **the minors of a block diagonal matrix are the products of matching-size minors of the blocks**. Here $xI - A$ is the direct sum of the blocks $xI_k - J_k(0)$, each upper triangular with x on the diagonal and -1 on the superdiagonal.

- (a) $xI - A = N \oplus N$ with $N = \begin{pmatrix} x & -1 \\ 0 & x \end{pmatrix}$, whose minors are the entries $x, -1, 0$ for $k = 1$ and x^2 for $k = 2$. By the remark, the minors of $N \oplus N$ are $\{x, -1, 0\}$ for $k = 1$; the products of two 1×1 minors together with x^2 for $k = 2$; x^2 times a 1×1 minor for $k = 3$; and x^4 for $k = 4$. Hence

$$\Delta_1 = 1, \quad \Delta_2 = 1 \quad (\text{from } (-1)(-1) = 1), \quad \Delta_3 = x^2, \quad \Delta_4 = x^4,$$

so the Smith normal form is $\text{diag}(1, 1, x^2, x^2)$, the invariant factors are $x^2 \mid x^2$, $m_A = x^2$ is the last of them, and the Jordan form is $J_2(0) \oplus J_2(0)$.

- (b) Now $xI - A = N_3 \oplus (x)$ with $N_3 = \begin{pmatrix} x & -1 & 0 \\ 0 & x & -1 \\ 0 & 0 & x \end{pmatrix}$. The minors of N_3 are $\{x, -1, 0\}$ for $k = 1$; $\{x^2, -x, 1, 0\}$ for $k = 2$, the value 1 coming from rows $\{1, 2\}$ and columns $\{2, 3\}$; and x^3 for $k = 3$. By the remark, the $k \times k$ minors of $xI - A$ are those of N_3 together with x times its $(k-1) \times (k-1)$ minors, so

$$\Delta_1 = 1, \quad \Delta_2 = \gcd(x^2, -x, 1, x \cdot x, -x) = 1, \quad \Delta_3 = \gcd(x^3, x \cdot x^2, -x \cdot x, x \cdot 1) = x, \quad \Delta_4 = x^4.$$

The Smith normal form is $\text{diag}(1, 1, x, x^3)$: the invariant factors are $x \mid x^3$, $m_A = x^3$ is the last of them, and the Jordan form is $J_3(0) \oplus J_1(0)$. Compared with (a) the determinants agree but Δ_3 does not, which is exactly why the two matrices are not similar.

- (c) $xI - A = \begin{pmatrix} x-3 & -1 \\ 0 & x-3 \end{pmatrix}$. Then $\Delta_1 = 1$ and $\Delta_2 = (x-3)^2$, so the Smith normal form is $\text{diag}(1, (x-3)^2)$, the only invariant factor is $(x-3)^2 = m_A = p_A$, and the Jordan form is $J_2(3)$. Consistently, A is a companion matrix up to similarity, since $m_A = p_A$ (Theorem 9.5.6).

Exercise B3: Deciding similarity.

Determine whether each pair is similar over the stated field. Justify your answer using Theorem 9.7.5.

- (a) $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ over $\mathbb{Q}$.
- (b) $A = \begin{pmatrix} 2 & 1 & -2 \\ 1 & 2 & -2 \\ 1 & 1 & -1 \end{pmatrix}$ and $B = J_2(1) \oplus J_1(1)$ over $\mathbb{Q}$.
- (c) $A = \begin{pmatrix} 0 & -2 \\ 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & -1 \\ 2 & 0 \end{pmatrix}$ over $\mathbb{Q}$.

Solution.

- (a) Similar. Here $B = A^\top$, so [Corollary 9.7.7](#) applies. (Explicitly, $\Delta_1 = 1$ and $\Delta_2 = x^2 + 1$ for both, so both Smith normal forms are $\text{diag}(1, x^2 + 1)$.)
- (b) Similar. By [Example \(A Smith Normal Form Computed by Hand\)](#) the Smith normal form of $xI - A$ is $\text{diag}(1, x - 1, (x - 1)^2)$. For B , $xI - B = \begin{pmatrix} x - 1 & -1 \\ 0 & x - 1 \end{pmatrix} \oplus (x - 1)$. Its entry -1 gives $\Delta_1 = 1$. By the block-minor rule in the solution to [Exercise \(B2: Invariant factors from determinantal divisors\)](#), the 2×2 minors are $(x - 1)^2$, from the 2×2 block, together with the products of a 1×1 minor of each block, that is $(x - 1)^2$, $-(x - 1)$ and 0 ; so $\Delta_2 = x - 1$. Finally $\Delta_3 = \det(xI - B) = (x - 1)^3$. The Smith normal forms agree, so $xI - A \approx xI - B$ and $A \sim B$.
- (c) Similar. Here

$$xI - A = \begin{pmatrix} x & 2 \\ -1 & x \end{pmatrix}, \quad xI - B = \begin{pmatrix} x & 1 \\ -2 & x \end{pmatrix},$$

and each has an entry that is a non-zero constant, so $\Delta_1 = 1$ in both cases; and $\Delta_2 = \det(xI - A) = x^2 + 2 = \det(xI - B)$. So both Smith normal forms are $\text{diag}(1, x^2 + 2)$, hence $xI - A \approx xI - B$ and $A \sim B$ by [Theorem 9.7.5](#). Note that $x^2 + 2$ is irreducible over $\mathbb{Q}$, so no Jordan form exists over $\mathbb{Q}$ and [Exercise \(C1: A matrix is similar to its transpose\)](#) would not have decided this pair. (Here $B \neq A^\top$, so [Corollary 9.7.7](#) does not apply directly; the Smith forms settle it anyway.)

9.7.6.3 C. Going deeper

Exercise C1: The same algorithm over the integers.

The proof of [Theorem 9.7.3](#) used only one property of $F[x]$: division with remainder, where the “size” of a non-zero element is its degree. The integers $\mathbb{Z}$ have the same property, with $|a|$ in place of $\deg$.

- (a) State the version of [Theorem 9.7.3](#) for $M \in M_n(\mathbb{Z})$ that the same proof gives, saying what replaces “monic”.
- (b) Compute the Smith normal form over $\mathbb{Z}$ of $M = \begin{pmatrix} 2 & 4 \\ 6 & 4 \end{pmatrix}$, naming the operations.
- (c) Every finite abelian group G with n generators is $\mathbb{Z}^n$ modulo the column span of some $M \in M_n(\mathbb{Z})$, the matrix of relations, exactly as V was $F[x]^n$ modulo the columns of $xI - A$. Using (b), identify the group with relation matrix M as a product of cyclic groups.

Hint: for (c), a diagonal relation matrix $\text{diag}(d_1, \dots, d_n)$ presents the product of the

cyclic groups $\mathbb{Z}/d_i\mathbb{Z}$.

Solution.

- (a) Every $M \in M_n(\mathbb{Z})$ can be brought by integer elementary operations — swap two rows or columns, multiply a row or column by ± 1 (the units of $\mathbb{Z}$), add an **integer** multiple of one row or column to another — to a diagonal matrix $\text{diag}(d_1, \dots, d_n)$ with every $d_i \geq 0$ and $d_i \mid d_{i+1}$, and this diagonal is unique. “Monic” is replaced by “non-negative”, which is the normalization available because the units of $\mathbb{Z}$ are ± 1 . Uniqueness again follows from the determinantal divisors Δ_k , now non-negative integer gcds of the $k \times k$ minors, using the Cauchy–Binet formula over the commutative ring $\mathbb{Z}$.
- (b) The least absolute value among the entries is 2, already in the corner. It divides 4 and 6. Subtract 2 times column 1 from column 2, and 3 times row 1 from row 2:

$$\begin{pmatrix} 2 & 4 \\ 6 & 4 \end{pmatrix} \longrightarrow \begin{pmatrix} 2 & 0 \\ 6 & -8 \end{pmatrix} \longrightarrow \begin{pmatrix} 2 & 0 \\ 0 & -8 \end{pmatrix} \longrightarrow \begin{pmatrix} 2 & 0 \\ 0 & 8 \end{pmatrix},$$

the last step multiplying row 2 by -1 . Since $2 \mid 8$, the Smith normal form is $\text{diag}(2, 8)$. Check: $\Delta_1 = \gcd(2, 4, 6, 4) = 2$ and $\Delta_2 = |\det M| = |8 - 24| = 16 = 2 \cdot 8$.

- (c) By (b) and the hint, the group is $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$, of order $16 = |\det M|$. This is the structure theorem for finite abelian groups in miniature: the invariant divisors $d_1 \mid \dots \mid d_n$ of the relation matrix are the orders of the cyclic factors.

Exercise C2: Uniqueness by hand in the smallest case.

Let $M \in M_2(F[x])$ be non-zero.

- (a) Prove directly, without using determinantal divisors, that $\Delta_1(M)$ — the monic gcd of the four entries of M — is unchanged by each of the six elementary operations of [Definition 9.7.1](#).
- (b) Prove that $\det(UMV) = c \det M$ for a non-zero constant c whenever U, V are invertible over $F[x]$.
- (c) Deduce that if $\text{diag}(d_1, d_2) \approx \text{diag}(e_1, e_2)$ with all four entries monic or zero, $d_1 \mid d_2$ and $e_1 \mid e_2$, then $d_1 = e_1$ and $d_2 = e_2$.

Solution.

- (a) Write $d = \Delta_1(M)$, the monic gcd of the entries, that is, the monic generator of the ideal $\langle m_{11}, m_{12}, m_{21}, m_{22} \rangle$ (Theorem 5.2.2). A swap permutes the entries, so the ideal is unchanged. Scaling a row by $c \in F \setminus \{0\}$ replaces two generators g by cg , and $\langle cg \rangle = \langle g \rangle$ since c is a unit, so the ideal is unchanged. Adding h times row 1 to row 2 replaces m_{2j} by $m_{2j} + hm_{1j}$, which lies in the old ideal; so the new ideal is contained in the old one, and the operation is reversed by adding $-h$ times row 1 to row 2, which gives the reverse containment. The same three arguments apply to columns. Hence d is unchanged.
- (b) By multiplicativity of the determinant over the commutative ring $F[x]$ (Theorem 6.5.4 and the remark on commutative rings), $\det(UMV) = \det U \cdot \det M \cdot \det V$, and $\det U, \det V$ are non-zero constants by [Proposition 9.7.1](#). Take $c = \det U \det V$.

- (c) By [Proposition 9.7.2](#) and (a), Δ_1 is an invariant of equivalence as far as elementary operations go; and for a general equivalence $M' = U M V$ the argument of (a) applies verbatim with the generators of the ideal replaced using $m'_{ij} = \sum_{k,l} u_{ik} m_{kl} v_{lj} \in \langle m_{kl} \rangle$, and symmetrically. For the two diagonal matrices, $\Delta_1(\text{diag}(d_1, d_2)) = \gcd(d_1, d_2) = d_1$ since $d_1 \mid d_2$ and d_1 is monic (or $d_1 = d_2 = 0$); likewise $\Delta_1(\text{diag}(e_1, e_2)) = e_1$. Hence $d_1 = e_1$. By (b), $d_1 d_2 = c e_1 e_2$ for some non-zero constant c ; if $d_1 = e_1 \neq 0$ then $d_2 = c e_2$, and two monic polynomials (or two zeros) differing by a constant factor are equal, so $d_2 = e_2$. If $d_1 = e_1 = 0$, then $d_2 = e_2 = 0$ by the divisibility chain.